

May 22, 2026

forall:gitevals sync

Invited specbtaiba@gmail.com ellahoeppner@gmail.com Quinn Dougherty

Attachments  forall:gitevals sync

Meeting records  Transcript

Summary

The team aligned on implementing agent-based classification for repository analysis while establishing a sequential development strategy.

Project Scope and Strategy

The team established a 3-week timeline to generalize the scaffold across 4 repositories. CompCert was selected as the first priority for sequential development.

Automated Classification Transition

Consensus was reached to move from hardcoded heuristics to AI agents for commit analysis. The fiat-crypto repository will serve as the initial baseline for comparison.

Dataset and Deliverables

Plans were set to generate Hugging Face draft posts using JSON lines data structure. The initial focus targets Supervised Fine-Tuning datasets for future repository research.

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Decisions

ALIGNED

- **Sequential repository generalization strategy adopted** The team decided to generalize the scaffold subdirectory sequentially across repositories rather than attempting multi-shot generalization.
- **Agent-based behavioral categorization adopted** The project will shift from hardcoded behavioral categories to an agent-based approach that dynamically generates categories based on repository data.
- **Agent-based fiat crypto replication prioritized** The team will first strip out hardcoded logic and use an agent to replicate existing fiat crypto work before generalizing the approach to other repositories.
- **SFT dataset format established** Hugging Face draft posts will be structured as SFT datasets in a JSON lines format, including repository metadata and before/after code examples.

We've **updated the Decisions section** using your feedback.

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Next steps

- ☐ [Quinn Dougherty] Discuss Fiat Crypto: Clarify if the browser-based code utilizes the Fiat Crypto toolchain by consulting with Jason.
- ☐ [Quinn Dougherty] Contact BAE Colleague: Inquire about the Isabelle language server protocol implementation used by the researcher at BAE.
- ☐ [Quinn Dougherty] Map Compser Repo: Identify subdirectories and code regions in the Compser repository and communicate findings to the team.
- ☐ [Taiba Abid] Study Compser: Review the repository structure to prepare for subsequent data generation tasks.

- ☐ [Taiba Abid] Refactor Scaffold: Strip out hardcoded logic in the scaffold repository and submit a pull request to enable agent-based generalization.
- ☐ [Ella Hoepfner] Strategize CompCert: Determine the strategy for implementing repository-wide scaffolding for CompCert.
- ☐ [Quinn Dougherty] Manage Artifacts: Clean up the repository and define a sustainable strategy for managing artifacts.
- ☐ [Quinn Dougherty] Evaluate Tracking Tools: Consult with Claude to assess the suitability of using Weights and Biases for project monitoring.

Details

- **Project Status and Initial Artifacts:** Quinn Dougherty reviews Taiba Abid's recent write-up, characterizing it as a solid intermediate artifact that effectively pulls the technical details together, though acknowledging the need for further experiments to fully address the philosophy of AI. Taiba Abid proposes addressing existing pipeline bugs and conducting additional experiments before separating the philosophy-related discussions into a distinct post ([00:00:00](#)).
- **Project Scope and Repository Targets:** Quinn Dougherty formally introduces Ella Hoepfner to the team and outlines a three-week timeline to prepare Hugging Face post versions for four specific repositories: fiat-crypto, brick, CompCert, and seL4/L4V ([00:03:33](#)). The central task is to replicate the scaffold subdirectory across these repositories. Quinn Dougherty emphasizes the significance of fiat-crypto, which serves as the cryptography framework for TLS and HTTPS, and mentions the potential to leverage an existing relationship with Greg Malica for support with the brick repository ([00:04:54](#)).
- **Technical Strategy and Capacity:** The team discusses the requirements for generalization, including the necessity of dockerized toolchains to manage compilation for the selected repositories. Regarding the approach to generalizing the scaffold subdirectory, Ella Hoepfner suggests a sequential strategy rather than attempting a simultaneous generalization, given the team's capacity constraints; Taiba Abid works approximately 20 hours, while Ella Hoepfner splits time between multiple projects ([00:07:26](#)). The team agrees to prioritize the CompCert repository first, noting that it may be more complex due to its multiple submodules and distinct subdirectories ([00:08:42](#)) ([00:11:59](#)).

- **Methodology for Git History Mining:** The core research methodology involves git history mining to analyze the life cycle of completed proofs by tracking commit messages and diffs. Ella Hoepfner clarifies that the objective is to track proofs as they evolve throughout the code base ([00:09:45](#)) ([00:14:29](#)). Quinn Dougherty explains that current heuristics classify commits into specific behavioral categories—proof add, proof modify, proof optimize, and proof refactor—to interpret the commit messages ([00:15:30](#)).
- **Transition to Agent-Based Classification:** The team discusses moving away from manually hardcoded heuristics toward using AI agents for automated judgment calls ([00:17:37](#)). Taiba Abid notes that previous categorization efforts were manually organized and insufficient for scalable Reinforcement Learning curriculum design ([00:15:30](#)) ([00:19:56](#)). The team reaches a consensus to remove these hardcoded constraints and employ an agent or Large Language Model to identify patterns and perform classifications automatically, starting with the fiat-crypto repository as a baseline for comparison ([00:18:37](#)) ([00:24:31](#)).
- **Implementation Plan and Assignments:** Taiba Abid will take the lead on refactoring the scaffold by stripping out hardcoded elements and implementing the new agent-based workflow using fiat-crypto ([00:28:21](#)). Ella Hoepfner will focus on mapping out the CompCert code base to identify specific subregions for the next stage of the project ([00:29:27](#)). Quinn Dougherty commits to managing artifact cleanup and addressing the challenges associated with repository artifact management ([00:30:44](#)).
- **Future Deliverables and Dataset Structure:** The team plans to produce minimal Hugging Face draft posts for each repository to share with colleagues ([00:25:43](#)). These datasets will be structured as JSON lines files containing metadata and before-and-after commit data, focusing initially on Supervised Fine-Tuning data sets before potentially expanding into Reinforcement Learning and curriculum design in the future ([00:27:08](#)).

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